

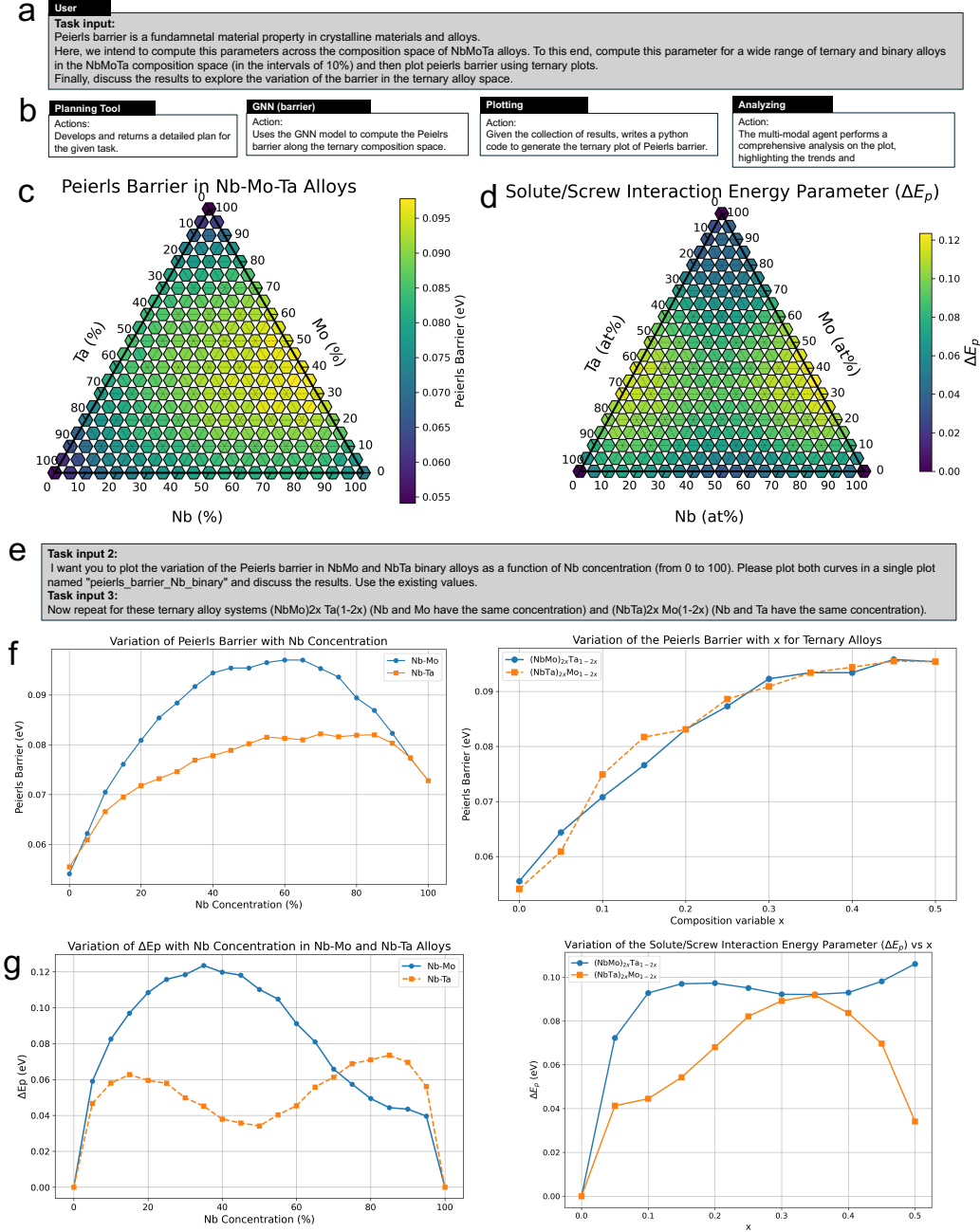


Figure 6: Overview of the multi-agent collaboration to explore the variation of the Peierls barrier and solute/screw interaction energy parameter in binary and ternary alloys. (a) User query requesting exploration of the Peierls barrier across a wide range of compositions, with a similar query repeated for the solute/screw interaction energy parameter. (b) Workflow detailing the computations and analyses performed by the multi-agent system. (c) and (d) Ternary plots displaying the Peierls barrier and the solute/screw interaction energy parameter, respectively, across the compositional space generated by the multi-agent system. (e) Follow-up tasks requested by the user, involving additional data plotting and analysis. (f) and (g) Plots illustrating the variation of the Peierls barrier and ΔE_p , respectively, with solute concentration for binary and ternary alloys, generated by the multi-agent model in response to the follow-up tasks.